



PLUNGER PUMP



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Introduction to Pumps

Definition: A hydraulic machine that converts *Mechanical Energy* (from a motor) into *Hydraulic Energy* (Pressure Energy).

Primary Functions:

- Transfer liquid from a low-pressure zone to a high-pressure zone.
- Lift liquid from a lower level to a higher level.
- Circulate liquid against system resistance.



Head (H)

Height/Pressure



Discharge (Q)

Flow Rate

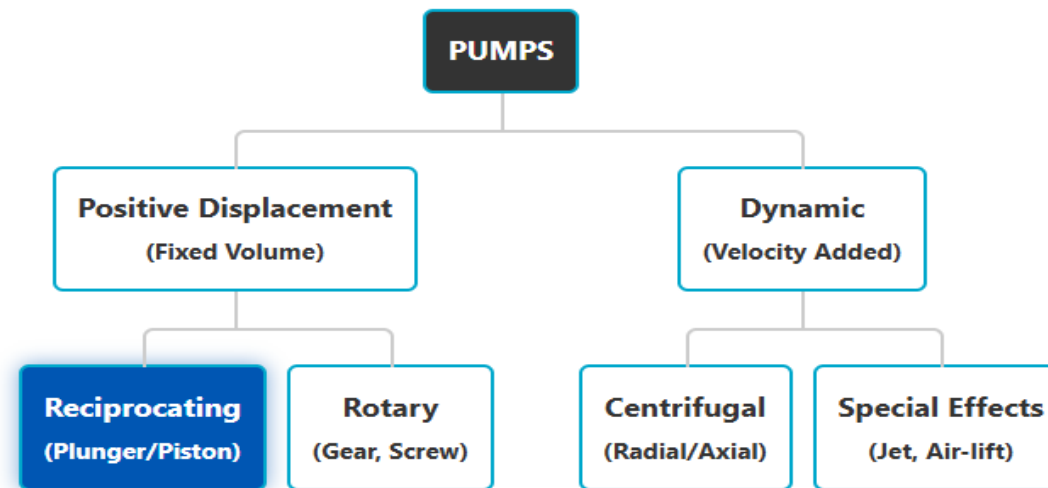


Power (P)

Energy Consumption

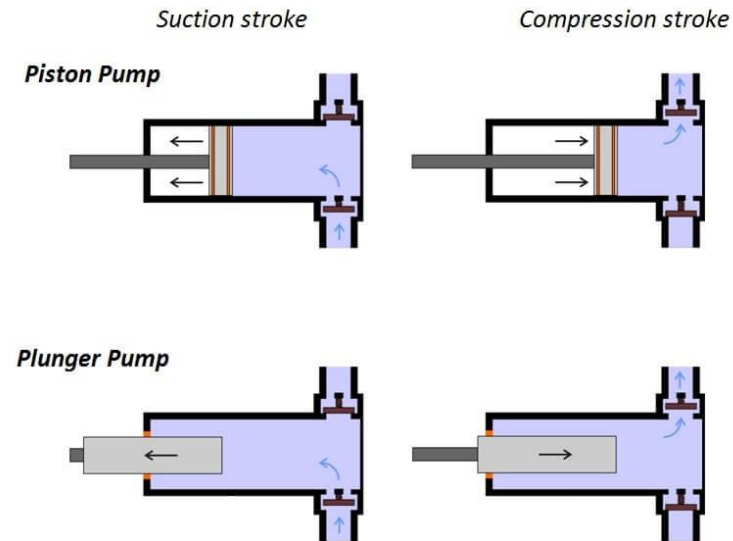
Classification of Pumps

Pumps are broadly classified based on how they transfer energy to the fluid.



Plunger Pump vs. Piston Pump

While both are reciprocating pumps, the key difference lies in the **sealing mechanism**.



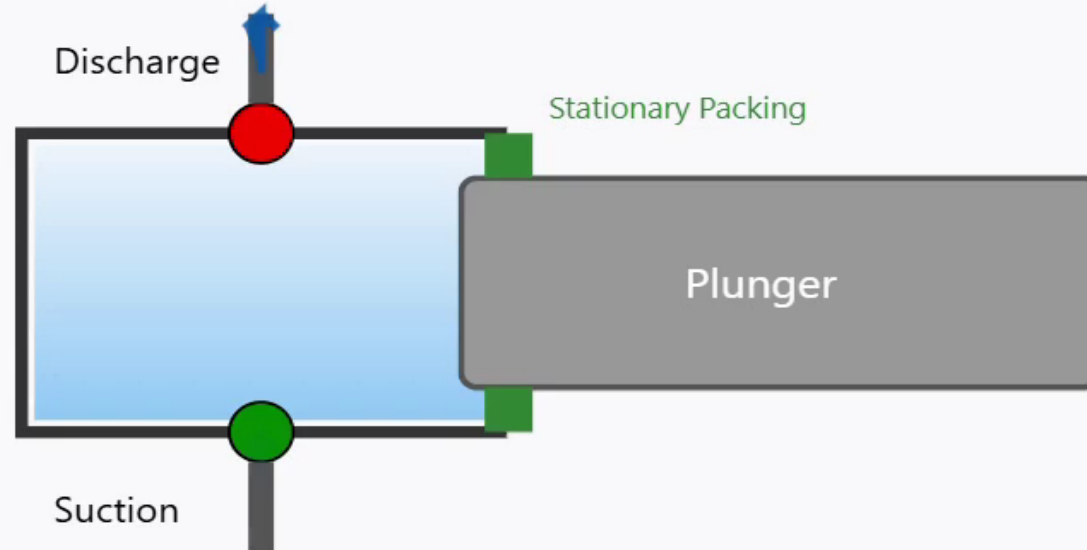
Source : <https://hosturl.link/GSvspq>

Feature	Plunger Pump
Pressure Capability	Extremely High (> 1000 bar)
Sealing	Stationary Packing (Easy to maintain)
Component	Solid, smooth plunger (Ceramic/Steel)

Pumps For Groundwater Lifting

Working Principle

The operation consists of two strokes: Suction and Discharge.



- 1.Suction Stroke (Plunger moves Right):** Vacuum created. Inlet valve opens (Green). Outlet valve closes (Red). Fluid enters.
- 2.Discharge Stroke (Plunger moves Left):** Volume decreases. Pressure rises. Inlet valve closes. Outlet valve opens. Fluid forced out.

Construction Features

1. The Plunger

Unlike a piston, the plunger is a solid cylinder. It is often made of **Ceramic** or high-grade **Stainless Steel** to resist wear and corrosion.

2. Stationary Packing

The seal does not move. This allows for tighter sealing arrangements, enabling the pump to reach pressures up to **20,000 PSI**.

3. Valves

Spring-loaded check valves (ball or poppet type) ensure uni-directional flow.

4. Drive Mechanism

Powered by an electric motor or diesel engine connected via a crankshaft to convert rotary motion to reciprocating motion.

Pumps For Groundwater Lifting

Suitability: When to use it?

High Pressure, Low Flow

Ideal for applications needing massive pressure where volume is secondary.

Viscous Fluids

Handles heavy oils, sludge, and slurries better than centrifugal pumps.

Metering / Dosing

Exact volume per stroke makes it perfect for chemical dosing (e.g., fertilizer injection).

Self-Priming

Can pump air/gas to clear lines, unlike standard centrifugal pumps.

Real-World Applications

- **High-Pressure Washing:** Car washes and industrial surface cleaning (The most common use).
- **Reverse Osmosis (RO):** Pushing salt water through membranes in desalination.
- **Oil & Gas:** Injecting methanol or glycol into wells at high pressure.
- **Water Jet Cutting:** Using water to cut steel/stone.

Advantages vs. Disadvantages

Advantages

- Capable of very high discharge pressures.
- High hydraulic efficiency.
- Constant flow rate (Positive Displacement).
- Self-priming capability.

Disadvantages

- **Pulsating Flow:** Requires dampeners.
- **Maintenance:** Valves and packing need monitoring.
- **Size/Weight:** Heavy and bulky.
- Higher initial cost than centrifugal pumps.

Conclusion:

"The Plunger Pump is the **Powerhouse of Pressure**. It sacrifices smooth flow for raw force and precision."

References

- Bansal, R. K. (2005). *A textbook of fluid mechanics and hydraulic machines*. Laxmi Publications.
- Grundfos. (2023). *The pump handbook*. Grundfos Management A/S.
- Modi, P. N., & Seth, S. M. (2002). *Hydraulics and fluid mechanics including hydraulics machines*. Standard Book House.
- Pump School. (n.d.). *Plunger vs piston pumps*.
- White, F. M. (2011). *Fluid mechanics* (7th ed.). McGraw-Hill Education.